

Phase 1 Report: Decision Trees and Gradient Boosting

Museum Indoor/Outdoor Binary Classification

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Abstract—This report presents Phase 1 of the COMP6721 course project, focusing on machine learning approaches to binary image classification of museum indoor and outdoor scenes using the MIT Places dataset. Three models are implemented and evaluated: a supervised Decision Tree, a Gradient Boosting ensemble, and a semi-supervised Decision Tree using iterative pseudo-labeling. Images are preprocessed as 32×32 grayscale pixel vectors. Results show Gradient Boosting ($F1=0.7158$) outperforms the best Decision Tree ($F1=0.6564$), while the semi-supervised approach achieves $F1=0.6231$ using only 20% labeled data initially. Phase 2 will extend this work with CNN-based deep learning models.

Index Terms—Decision Tree, Gradient Boosting, Semi-Supervised Learning, Image Classification, Places MIT Dataset

I. INTRODUCTION AND PROBLEM STATEMENT

Classification of images is a fundamental computer vision problem with use cases ranging from autonomous navigation to digital content management, museum digitization, and smart environments. This project tackles the binary image classification problem of separating indoor and outdoor museum pictures in the Places MIT dataset [?]. Being able to correctly classify the indoor or outdoor nature of a picture is helpful for applications that adapt their functionality depending on the spatial setting, such as augmented reality applications, surveillance, or robotic navigation.

The key difficulty lies in the potential visual similarity between indoor and outdoor museum images, where both categories may share similar architectural elements and color palettes. Additional challenges include class imbalance (8,119 indoor vs. 4,221 outdoor training images), a high-dimensional input space from raw pixel features, and the intrinsic limitations of Decision Trees in such settings.

Three machine learning models are presented in this report: a supervised Decision Tree, a Gradient Boosting ensemble, and a semi-supervised Decision Tree that utilizes pseudo-labeling to iteratively enlarge the set of labeled images. The pipeline consists of image preprocessing, feature extraction through 32×32 grayscale pixel flattening, hyperparameter tuning, and performance evaluation using standard classification metrics.

II. PROPOSED METHODOLOGY

A. Dataset

Data was obtained from the MIT Places Database [?]. The training dataset includes 10,000 images (5,000 per class after balancing), with the validation dataset comprising 200 images (100 indoor, 100 outdoor). Images were converted to grayscale and resized to 32×32 pixels, producing 1,024-dimensional feature vectors. Pixel values were normalized to $[0, 1]$ by dividing by 255.

B. Feature Extraction

Images were converted to grayscale to reduce dimensionality, then resized to 32×32 pixels using the Lanczos resampling filter. The resulting image was flattened into a 1,024-element feature vector. This representation does not demand extensive feature engineering and is well-suited for Decision Tree-based classifiers.

C. Supervised Decision Tree

Three Decision Tree classifiers (`DecisionTreeClassifier` from `scikit-learn` [?]) were trained with `max_depth` values of 5, 10, and 20, using Gini impurity and a minimum of 10 samples per split. The depth hyperparameter controls the bias-variance trade-off: shallow trees underfit the data, while very deep trees may overfit. All models were evaluated on the validation set.

D. Gradient Boosting

Gradient Boosting (`GradientBoostingClassifier`) was applied with `n_estimators` values of 50, 100, and 200, a fixed `max_depth` of 3 per weak learner, and a learning rate of 0.1. Boosting builds an ensemble of weak learners sequentially, each correcting the errors of its predecessor, and is expected to outperform single Decision Trees by reducing both bias and variance [?].

E. Semi-Supervised Decision Tree

To simulate a real-world scenario where only a fraction of the data is labeled, 20% of training samples (2,000 images) were randomly selected as the labeled set, while the remaining 80% (8,000 images) were treated as unlabeled. A Decision Tree ($\text{max_depth}=10$) was trained on the labeled set, and pseudo-labels were generated for the unlabeled data. Samples with predicted probability ≥ 0.85 (outdoor) or ≤ 0.15 (indoor) were iteratively added to the labeled pool for up to 5 rounds, following the self-training paradigm [?].

III. RESULTS AND DISCUSSION

A. Supervised Decision Tree Results

Table ?? presents the performance of all supervised models on the validation set.

TABLE I
SUPERVISED MODEL PERFORMANCE ON VALIDATION SET

Model	Acc.	Prec.	Recall	F1
DT (depth=5)	0.6700	0.6977	0.6000	0.6452
DT (depth=10)	0.6650	0.6737	0.6400	0.6564
DT (depth=20)	0.6450	0.6465	0.6400	0.6432
GB (n=50)	0.7300	0.7556	0.6800	0.7158
GB (n=100)	0.7350	0.7765	0.6600	0.7135
GB (n=200)	0.7350	0.7831	0.6500	0.7104

Among Decision Tree variants, $\text{depth}=10$ achieved the best F1-score of 0.6564. The $\text{depth}=5$ tree underfits the data ($\text{F1}=0.6452$), while $\text{depth}=20$ shows slight overfitting ($\text{F1}=0.6432$). Gradient Boosting consistently outperformed single Decision Trees across all metrics, with GB (n=50) achieving the best F1 of 0.7158 and GB (n=100) achieving the highest accuracy of 0.7350. Increasing estimators beyond 100 yields diminishing returns, suggesting model convergence near $n=100$.

B. Semi-Supervised Decision Tree Results

Table ?? summarizes the semi-supervised learning progression over 4 iterations.

TABLE II
SEMI-SUPERVISED DT PERFORMANCE OVER ITERATIONS

Iteration	Acc.	Prec.	Recall	F1
Iter 1 (20% labelled)	~ 0.60	—	—	~ 0.60
Iter 2 (+552 pseudo)	0.6100	0.6146	0.5900	0.6020
Iter 3 (+19 pseudo)	0.6150	0.6162	0.6100	0.6131
Iter 4 (0 added)	0.6250	0.6263	0.6200	0.6231

Starting with only 20% labeled data, the semi-supervised model achieved an F1 of approximately 0.60 in the first iteration, lower than the fully supervised DT (0.6564) as expected. Iteration 2 added 552 high-confidence pseudo-labels, and F1 improved marginally. By iteration 3, only 19 additional samples met the confidence threshold, and iteration 4 found zero confident predictions, causing early stopping. The final semi-supervised F1 of 0.6231 remains below the fully

supervised counterpart, reflecting limited pseudo-label quality arising from grayscale 32×32 pixel features rather than an algorithmic failure of the self-training paradigm.

C. Confusion Matrix Analysis

All models showed difficulty classifying outdoor images, exhibiting high false-negative rates. This is attributable to the class imbalance in the training set (8,119 indoor vs. 4,221 outdoor), causing the models to develop a bias toward the majority indoor class. Gradient Boosting partially alleviates this bias compared to single Decision Trees, as evidenced by higher recall values across all ensemble configurations.

IV. FUTURE IMPROVEMENTS

Several directions could improve model performance in Phase 2. First, replacing raw pixel features with engineered descriptors such as HOG or RGB color histograms would provide more semantically meaningful representations. Second, increasing image resolution to 64×64 or 128×128 pixels could preserve more discriminative spatial detail. Third, addressing class imbalance through oversampling (e.g., SMOTE) or class-weighted training could improve outdoor recall. Finally, the CNN-based models planned for Phase 2 (custom CNN and ResNet transfer learning) are expected to substantially outperform pixel-based Decision Trees by learning hierarchical feature representations directly from image data.

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